Value putting:

i) No. of variables > no. of eqns.

ii) Formula asked { v → v, v → c }

$\theta \rightarrow 90^\circ/0^\circ,\ 45^\circ,\ 30^\circ/60^\circ$

$\frac{\theta}{2}, 2\theta \rightarrow 60, 30, 15$

$\sec = \operatorname{cosec} = \sqrt{2}$

$\sin = \cos = \frac{1}{\sqrt{2}}$

$\tan = \cot = 1$

iii) Misc.
- Putting all angles equal. (Symmetrical)
- Substituting complementary values.

1) $(a+b)^2 = ?$

a) a^2+b^2+2ab

ii) $\frac{(a+b)^2+(a-b)^2}{(a^2+b^2)} = ?$

a) 2

$A = 45$

① The simplified value of $(secA - cosA)^2 + (cosecA - sinA)^2 - (cotA - tanA)^2$

$(secA - cosA)^2 + (cosecA - sinA)^2 - (cotA - tanA)^2$ को सरल करें:

a) 0 b) $\frac{1}{2}$

~~c) 1~~ d) 2

$\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)$

$\frac{1}{2} + \frac{1}{2} - 0 = 1$

coaching center

$\theta = 45$

② The value of $(1 + cot\theta - cosec\theta)(1 + tan\theta + sec\theta)$ is equal to

$(1 + cot\theta - cosec\theta)(1 + tan\theta + sec\theta)$ का मान है:

a) 1 b) 2

c) 0 d) -1

$(2-\sqrt{2})(2+\sqrt{2}) = 2$

coaching center

$\alpha = 45$

$$\frac{9}{2} + \frac{9}{2} = k + 2$$

$$7 = k$$

③ If $(\sin\alpha + \operatorname{cosec}\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is

अगर $(\sin\alpha + \operatorname{cosec}\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, है, तो k का मान:

a) 1 b) 7

c) 3 d) 5

$$\sqrt{2} + \frac{1}{\sqrt{2}}$$

coaching center

$\theta = 0$

(4) Find the value of $3(\sin^4\theta + \cos^4\theta) + 2(\sin^6\theta + \cos^6\theta) + 12\sin^2\theta\cos^2\theta$

$3(\sin^4\theta + \cos^4\theta) + 2(\sin^6\theta + \cos^6\theta) + 12\sin^2\theta\cos^2\theta$ का मान क्या है|

a) 3 b) 2

c) 0 d) 5

$3+2$

e1 coaching center

$x = y = 45$

$9 - 8 = 1$

⑤ The simplified value of $(sec\ x\ sec\ y + tan\ x\ tan\ y)^2 - (sec\ x\ tan\ y + tan\ x\ sec\ y)^2$ $(sec\ x\ sec\ y + tan\ x\ tan\ y)^2 - (sec\ x\ tan\ y + tan\ x\ sec\ y)^2$ को सरल करें|

a) -1 b) 0

c) $sec^2\ x$ d) 1

e1 coaching center

⑥ What is the value of $\frac{[(\sin x+\sin y)(\sin x-\sin y)]}{(\cos x+\cos y)(\cos y-\cos x)}$?

$\frac{[(\sin x+\sin y)(\sin x-\sin y)]}{(\cos x+\cos y)(\cos y-\cos x)}$ का मान क्या है?

a) 0 ~~b) 1~~

c) -1 d) 2

$x=0$

$$\frac{-\sin^2}{-\sin^2} = \frac{}{\cos^2 y-1} = \frac{-\sin^2 y}{(1+\cos y)(\cos y-1)}$$

$y=0$

$$\frac{\sin^2 x}{(\cos x+1)(1-\cos x)} = \frac{\sin^2}{\sin^2}$$

coaching center

(7) What is the value of $\sin(B-C)\cos(A-D)+\sin(A-B)\cos(C-D)+\sin(C-A)\cos(B-D)$?

$\sin(B-C)\cos(A-D)+\sin(A-B)\cos(C-D)+\sin(C-A)\cos(B-D)$ का मान क्या है?

a) $\frac{3}{2}$ b) -3

c) 1 d) 0

$A=B=C=D$

$0+0+0=0$

coaching center

⑧ What is the value of $cos(90-B)\,sin(C-A) + sin(90+A)\,cos(B+C) - sin(90-C)\,cos(A+B)$?

$cos(90-B)\,sin(C-A) + sin(90+A)\,cos(B+C) - sin(90-C)\,cos(A+B)$ का मान क्या है?

a) 1 b) $sin(A+B-C)$

c) 0 d) $cos(B+C-A)$

$A=B=C$

$0+\cos A\cos 2A-\cos A\cos 2A$

$A=C=0$

$0+\cos B-\cos B=0$

$\sin B\sin(C-A)+\cos A\cos(B+C)-\cos C\cos(A+B)$

$A=C$

$0+\cos A(\cos A+B)-\cos A(\cos A+B)$

$=0$

coaching center

$x = y = z$

$$\frac{0 + \sin 2x + 2\sin x}{0 + \sin 2x + 2\sin x} = 1$$

⑨ **What is the value of**

$$\frac{[\sin(y-z)+\sin(y+z)+2\sin y]}{\sin(x-z)+\sin(x+z)+2\sin x}$$?

$$\frac{[\sin(y-z)+\sin(y+z)+2\sin y]}{\sin(x-z)+\sin(x+z)+2\sin x}$$ **का मान क्या है?**

a) $\cos x \sin y$

b) $\frac{\sin y}{\sin x}$

c) $\sin z$

d) $\sin x \tan y$

coaching center

SSC CGL 2017 Mains

$\theta = 0$

$$\frac{1+3}{1} = 4$$

10 What is the value of $[(Cos^3 2\theta + 3\,Cos\,2\theta) \div (Cos^6\,\theta - Sin^6\theta)]$?

$[(Cos^3 2\theta + 3\,Cos\,2\theta) \div (Cos^6\,\theta - Sin^6\theta)]$ का मान क्या है?

a) 0 b) 1

c) 4 d) 2

coaching center

⑪ $\frac{\cot\theta}{\cot\theta - \cot 3\theta} + \frac{\tan\theta}{\tan\theta - \tan 3\theta} = ?$

a) 1 b) -1

c) 2 d) 0

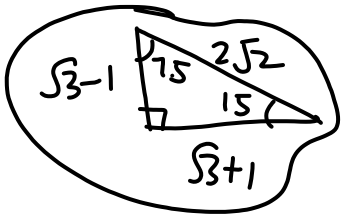
$\theta = 45$

$\frac{1}{1-(-1)} + \frac{1}{1-(-1)} = \frac{1}{2} + \frac{1}{2} = 1$

$\cot(90+45)$ $\quad$ $\tan(90+45)$

$= -\tan 45$ $\quad$ $= -\cot 45$

$= -1$ $\quad$ $= -1$



coaching center

$\theta = 45$

$$\frac{(\sqrt{2}-1)}{1} \times \frac{1}{\sqrt{2}-1} = 1$$

(12) The value of $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\text{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$ is:

$\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\text{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$ का मान क्या है?

a) 0 b) -1

c) 1 d) $\frac{1}{2}$

e1 coaching center

$A = 0$

$\frac{0}{\text{◯}} = 0$

$A = 45$

(13) $\frac{(2sinA)(1+sinA)}{1+sinA+cosA}$ is equal to:

$\frac{(2sinA)(1+sinA)}{1+sinA+cosA}$ का मान किसके बराबर है?

1 a) $1 + sinA - cosA$ 0

½ b) $1 - sinAcosA$ 1

1 c) $1 + cosA - sinA$ 2

3/2 d) $1 + sinAcosA$ 1

coaching center

(14) What is the value of $\sin(90° + 2A)[4 - \cos^2(90° - 2A)]$?
$\sin(90° + 2A)[4 - \cos^2(90° - 2A)]$ का मान क्या है?

a) $2(\cos^3 A - \sin^3 A)$
b) $2(\cos^3 A + \sin^3 A)$
c) $4(\cos^6 A + \sin^6 A)$
d) $4(\cos^6 A - \sin^6 A)$

$\cos 2A [4 - \sin^2 2A]$

$A = 0$

$1[4 - 0] = 4$

$A = 45$

$0[4 - 1] = 0$

coaching center

$\theta = 45$

$x = \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$y = \frac{1}{\sqrt{2}}$

Put these values in options & check,

a) $\frac{1}{2} + \frac{1}{2} + 3 = 1$ ✗

b) $\frac{1}{2} \times \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2} + 3\right) = 1$ ✓

(15) (HW) If $x = cosec\theta - \sin\theta$ and $y = \sec\theta - \cos\theta$, then the relation between x and y is

यदि $x = cosec\theta - \sin\theta$ और $y = \sec\theta - \cos\theta$ हो, तो x और y के बीच सम्बन्ध होगा

a) $x^2 + y^2 + 3 = 1$

b) $x^2y^2(x^2 + y^2 + 3) = 1$ ✓

c) $x^2(x^2 + y^2 - 5) = 1$

d) $y^2(x^2 + y^2 - 5) = 1$

e1 coaching center

$\theta = 45$

$p = 2 \;,\; q = \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$(p^2q)^{\frac{2}{3}} - (pq^2)^{2/3}$

$= \left(\left(4 \frac{1}{\sqrt{2}}\right)^2\right)^{\frac{1}{3}} - \left(2 \times \frac{1}{2}\right)^{2/3}$

$= 2 - 1 = 1$

(16) If $cot\,\theta + tan\,\theta = p$ and

(HW) $sec\,\theta - cos\,\theta = q$ then

$(p^2q)^{\frac{2}{3}} - (pq^2)^{\frac{2}{3}} = ?$

यदि $cot\,\theta + tan\,\theta = p$ और $sec\,\theta - cos\,\theta = q$ है, तो

$(p^2q)^{\frac{2}{3}} - (pq^2)^{\frac{2}{3}} = ?$

a) 0 b) 1

c) 2 d) 3

coaching center

CDS 2016 I

(17) If $\frac{\cos\alpha}{\sin\alpha+\cos\beta}+\frac{\cos\beta}{\sin\beta-\cos\alpha}=\frac{x}{\sin\alpha-\cos\beta}+\frac{\cos\beta}{\sin\beta+\cos\alpha}$ then 'x' is equal to:

यदि $\frac{\cos\alpha}{\sin\alpha+\cos\beta}+\frac{\cos\beta}{\sin\beta-\cos\alpha}=\frac{x}{\sin\alpha-\cos\beta}+\frac{\cos\beta}{\sin\beta+\cos\alpha}$ है, 'x' तो बराबर है:

a) $\cos\beta$ b) $\cos\alpha$

c) $\sin\beta$ d) $\sin\alpha$

$\beta=90$

$\frac{\cos\alpha}{\sin\alpha}=\frac{x}{\sin\alpha}+X$

$\int 0-0$

$\alpha\neq 0$

coaching center

$\theta = 30$

$\left[3 \times \frac{1}{\sqrt{3}}\right] \times \frac{1}{2}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)$

$= \sqrt{3} - \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

(18) What is the value of $\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\ \theta - tan\ \theta)$?

$\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\,\theta - tan\,\theta)$ का मान क्या है?

a) 0 b) 1

c) $cosec\ \theta$ d) $sec\ \theta$

coaching center

(19) If $sinA + sinB = C$ & $cosA + cosB = D$, then $sin(A+B) = ?$

यदि $sinA + sinB = C$ और $cosA + cosB = D$ है, तो $sin(A+B) = ?$

a) CD

b) $\frac{CD}{C^2+D^2}$

c) $\frac{C^2+D^2}{2CD}$

d) $\frac{2CD}{C^2+D^2}$

$A = B = 0$ ✗

$A = B = 45$

$C = \sqrt{2}$ $D = \sqrt{2}$

$Sin(A+B) = 1$

$A = B = 30$

$C = 1$, $D = \sqrt{3}$

$Sin(A+B) = \frac{\sqrt{3}}{2}$

coaching center

$\alpha+\beta=45$

$\frac{1}{2} \times \frac{1}{2} \times 2 \times 2 = 1$

(20) If $\alpha + \beta = 90°$ then the value of $(1 - \sin^2 \alpha)(1 - \cos^2 \alpha) \times (1 + \cot^2 \beta)(1 + \tan^2 \beta)$ is

अगर $\alpha + \beta = 90°$ है, तो $(1 - sin^2 \alpha)(1 - cos^2 \alpha) \times (1 + cot^2 \beta)(1 + tan^2 \beta)$ का मान:

a) 1 b) -1

c) 0 d) 2

e1 coaching center

$\angle A = \angle B = 45$

$2 + 2 - 2 \times 2 = 0$

21 HW

If $\angle A$ and $\angle B$ are complementary to each other, then the value of $\sec^2 A + \sec^2 B - \sec^2 A \sec^2 B$ is

अगर कोण $\angle A$ और $\angle B$ एक दुसरे के पूरक हैं, तो $sec^2 A + sec^2 B - sec^2 A sec^2 B = ?$

a) 1 b) -1

c) 2 d) 0

e1 coaching center

$\theta = 45$

$\left(\frac{1}{8} + \frac{1}{8} - 1\right)(1 + 1 + 2)$

$= -\frac{3}{4} \times 4 = -3$

22 HW The value of the expression $(cos^6\theta + sin^6\theta - 1)(tan^2\theta + cot^2\theta + 2)$ is:

व्यंजक $(cos^6\theta + sin^6\theta - 1)(tan^2\theta + cot^2\theta + 2)$ निम्न में से किसके बराबर है?

a) 0 b) -1

c) -3 d) 1

coaching center

(23) $tan\left(\frac{\pi}{4}+\theta\right)-tan\left(\frac{\pi}{4}-\theta\right)=?$

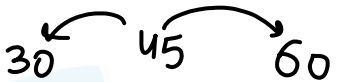
a) $2\ tan\ 2\theta$ 2/√3 b) $2\ cot\ 2\theta$ 2√3

c) $tan\ 2\theta$ 1/√3 d) $cot\ 2\theta$ √3

$\theta = 15$

$\tan(45+\theta) - \tan(45-\theta)$

$\sqrt{3} - \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$



coaching center

$\frac{3}{4} \times 3 - 2 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

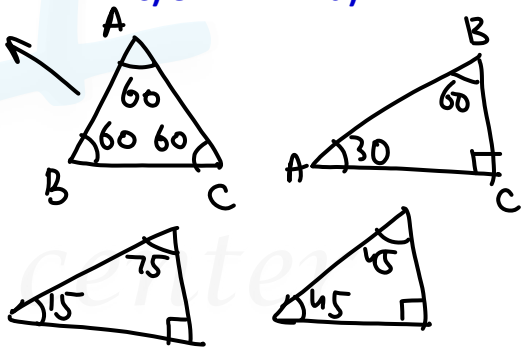
$= \frac{9}{4} - \frac{1}{4} = \frac{8}{4} = 2$

(24) If A, B, C are angles of a triangle, then $sin^2A + sin^2B + sin^2C - 2cosA\ cosB\ cosC = ?$

यदि A, B, C एक त्रिभुज के कोण हैं, तो $sin^2A + sin^2B + sin^2C - 2cosA\ cosB\ cosC = ?$

a) 1 b) 2

c) 3 d) 4



coaching center

(25) If $A+B+C=\frac{3\pi}{2}=270$ then $cos2A+cos2B+cos2C=?$

यदि $A+B+C=\frac{3\pi}{2}$ है, तो $cos2A+cos2B+cos2C=?$

a) $1-4\,sinA\,sinB\,sinC$

b) $1-sinA\,sinB\,sinC$

c) $1-2sinA\,sinB\,sinC$

d) $1-4cosA\,cosB\,cosC$

$90=C=B=A$

$-1-1-1=-3$

e1 coaching center